

**ΟΙΚΟΝΟΜΙΚΟ
ΠΑΝΕΠΙΣΤΗΜΙΟ
ΑΘΗΝΩΝ**



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Social Network Analysis of Christopher Paolini on Gephi

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Introduction

Christopher Paolini is an American and Italian author who is best known for “The Inheritance Cycle” series (“Christopher Paolini”, 2025) and has garnered a substantial following on social media platforms. The decision to analyse Paolini's network stems from his significant influence in the literary community, as well as myself, given that his novel “To Sleep in a Sea of Stars” is my favourite science fiction book.

His presence on Bluesky, a decentralized social media network, presents an intriguing opportunity to explore the structure and dynamics of his social network, which provides a distinct blend of professional and fan-based connections, making it an ideal case study for understanding the influence of an author within a digital space and the way literary communities form and function on social media platforms.

The analysis is conducted using the tool “Gephi” and its “Bluesky” plugin. The network created by fetching the data is a directed graph, in which the nodes represent individual accounts on Bluesky and the edges represent the followers of these accounts.

This analysis aims to achieve several objectives: to map out the topology of Paolini's network, identify key nodes that act as bridges or hubs, and explore the community structures within the network. By delving into these aspects, I hope to gain a deeper understanding of the network's overall connectivity, the role of influencers, and the nature of the connections within Paolini's digital presence in Bluesky.

Graphical Representation of the Network

An initial graphical representation of the network was created using the “ForceAtlas 2” layout. At first glance, it becomes evident that many users are connected to only a few others, positioning them on the periphery of the network. In contrast, there are several users who are connected to a larger number of individuals, which places them at the centre of the network.

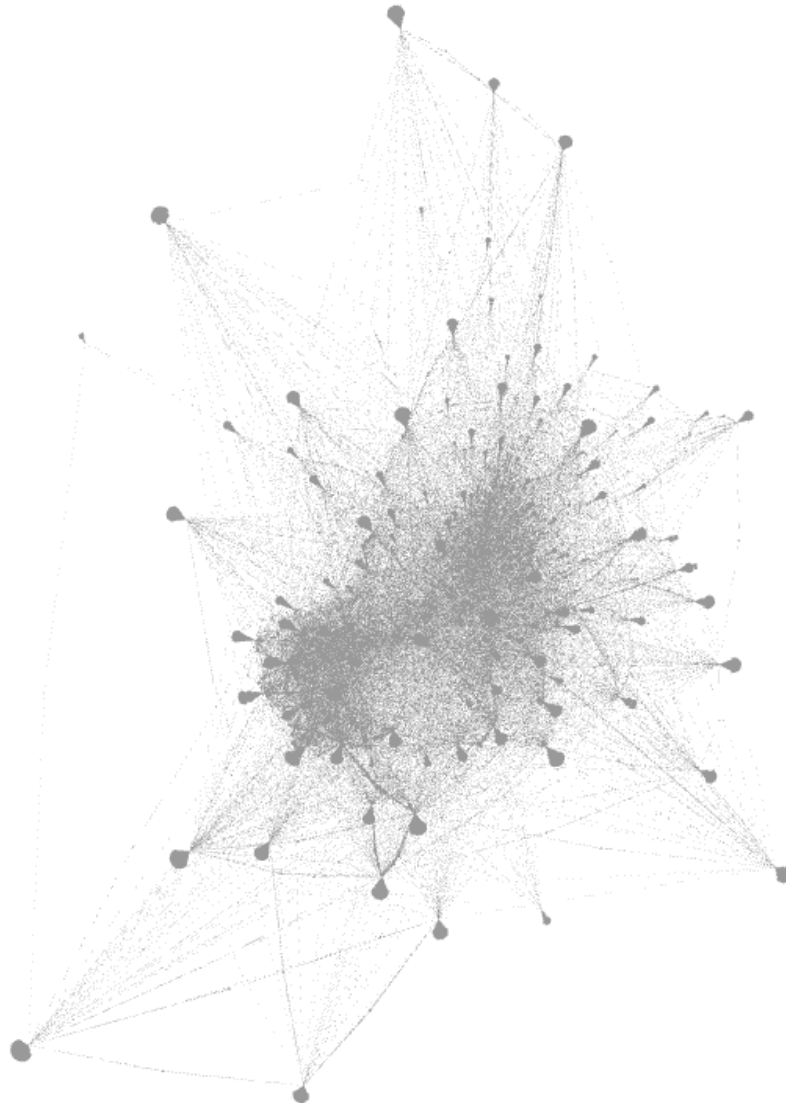


Figure 1: Graphical Representation of the Network

Additionally, after an addition in labels and a change in their colour and size based on their degree, the graphical representation changes. The colour palette used is , in which the smaller an account's degree the lighter its node's colour.

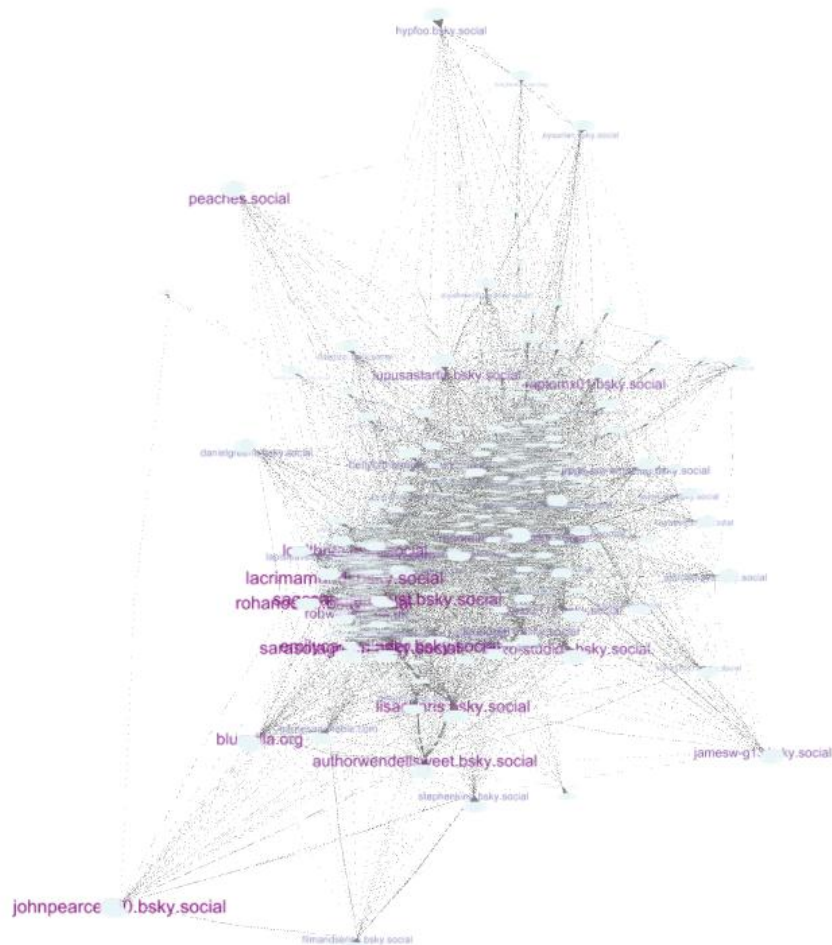


Figure 2: Coloured Graphical Representation of the Network

The network is evidently quite sparse, with numerous groups scattered around the main component. Most users exhibit low degree values, while a select few have significantly higher degrees, marking them as the core nodes of the network. These high-degree nodes act as central hubs, holding the network together and facilitating connectivity across different parts.

Basic Topological Properties

Numbers of Nodes and Edges

The graph is directed, having 59.052 nodes and 86.089 edges. As mentioned in the introduction, the nodes represent individual accounts on Bluesky, including Christopher Paolini and others connected to him. The edges, on the other hand, represent the followers of these accounts. More specifically, a directed edge from node A to node B indicates user A following user B.

Network Diameter

The diameter of a network is the maximum shortest path in the network (Barabási, 2016). In the network, the diameter is 11, meaning that the maximum number of steps needed to travel from one node to another, in the most efficient way, is 11. This suggests that even the most distant nodes in the network are relatively connected, requiring at most 11 steps to connect them.

Average Path Length

The average path length is the average distance between all pairs of nodes in the network (Barabási, 2016). In the network, it is approximately 5,44 (more accurately 5,436479668763233). This is the average number of steps along the shortest paths for all possible pairs of nodes in the graph. Thus, indicating that, on average, any node can be reached from any other node in about 5,44 steps, which means that the network is relatively well-connected.

However, the average path length can be somewhat misleading when assessing the overall connectivity of the network. A few users with a large number of followers form numerous small groups, which significantly shortens the average path length. This gives the impression of a well-connected network, despite the actual presence of many sparsely connected components.

Component Measures

Connected Components

Two vertices are in the same weakly connected component if they are connected by one or more paths through the network, where paths are allowed to go either way along any edge (Newman, 2018). When a network component's size grows to N , which is the number of nodes, the component is instead called a giant component (Newman, 2018). In this network, the number of weakly connected components is 1, which means that all the nodes in the network are connected in some way if you disregard the direction of the edges. Thus, forming a single giant component where every user is indirectly connected to every other user, even if only through a chain of connections.

On the other hand, a strongly connected component is a maximal subset of vertices such that there is a directed path in both directions between every pair in the subset (Newman, 2018). The number of strongly connected components is 50.223, suggesting that there are many smaller groups within the network where users are tightly interconnected in both directions. Most of these components are likely small, possibly individual users or small groups, which indicates that the network has a complex structure with many isolated or semi-isolated clusters when considering the direction of connections.

These metrics suggest that the network is densely connected in terms of overall reachability but consists of many smaller, isolated clusters, rendering it quite sparse.

Component Size Distribution

In this case, as expected, the component size distribution graph indicates that there is a single giant component with 59.052 nodes. The size distribution graph doesn't show much variation because the entire network is dominated by this giant component.

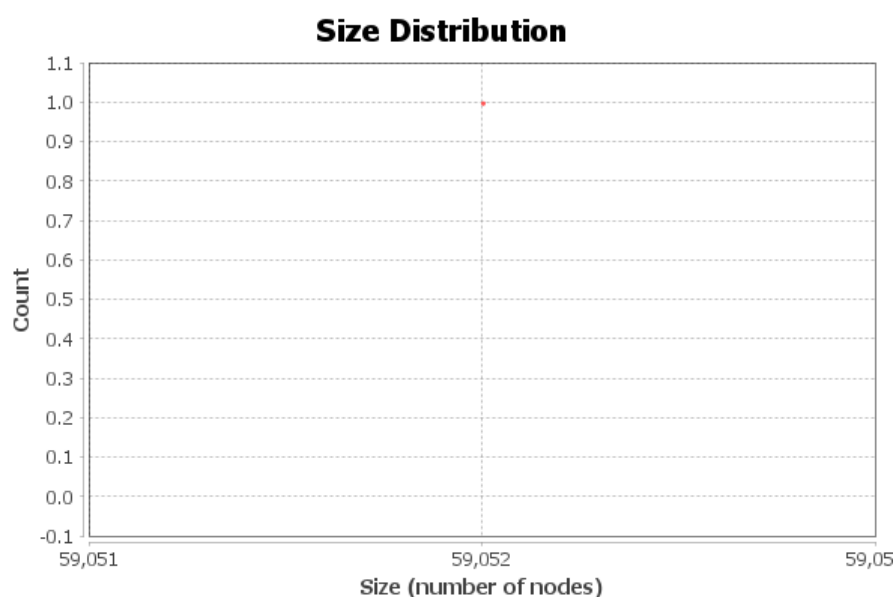


Figure 3: Component Size Distribution

Degree Components

Degree Distribution

A key property of each node is its degree, representing the number of links it has to other nodes (Barabási, 2016). Degree is sometimes called degree centrality in the social network's literature, to emphasize its use as a centrality measure (Newman, 2018). The average degree in the network is 1,458. Consequently, most nodes have a small number of connections, while a few may have many, forming a long tail in the distribution. This is typical in social networks, where a few users are very central with many connections, creating hubs, while most users are peripheral with few connections.

In this network, the average degree and the average weighted degree are the same, since the weight of all edges is equal to 1.

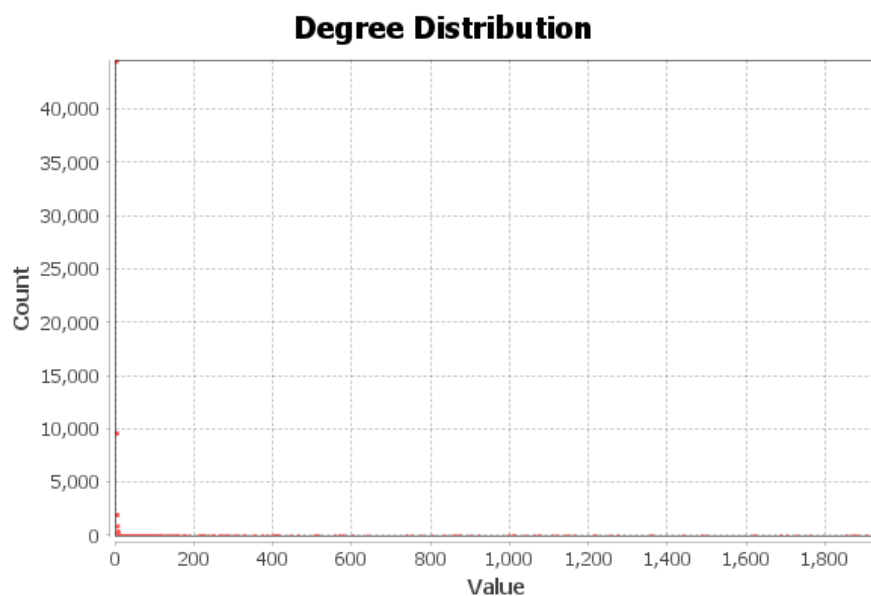


Figure 4: Degree Distribution

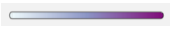
To better understand the degree in the network, this is the visualization of the network in which the colour and the size of the nodes is determined by the degree of the node using the layout “Fruchterman Reingold” and this colour scheme , in which the smaller an account's degree the lighter its node's colour:



Figure 5: Degree Distribution in the Network

The top 3 users in terms of degree are:

Label	Description	Weighted In-Degree	Weighted Out-Degree	Weighted Degree ▾
johnpearce650.bsky.social	Writer and environmentalist. Find me on Medium: http	929.0	1006.0	1935.0
lacrimamundi.bsky.social	Ardent Eshteross admirer. Writer, reader & reviewer. ...	917.0	986.0	1903.0
rohanoduill.bsky.social	Storyteller, chef 🍳 archer 🏹 Irish 🍀 dyslexic. Reppe...	895.0	984.0	1879.0

Figure 6: Top Users in Degree Distribution

All 3 of these users, @johnpearce650.bsky.social, @lacrimamundi.bsky.social and @rohanoduill.bsky.social, are, as can be seen from their descriptions, writers, and are generally well-established within the network. Their high degrees, in-degrees, and out-degrees grant them central positions in the network.

In-degree Distribution

In-degree corresponds to the number of edges pointing inward to those vertices (Newman, 2018). In the network, this represents how many nodes have a certain number of followers. It is likely that the in-degree distribution is skewed, with many nodes having few followers, and a few nodes having many incoming connections. This indicates that a small group of profiles are highly influential or popular.

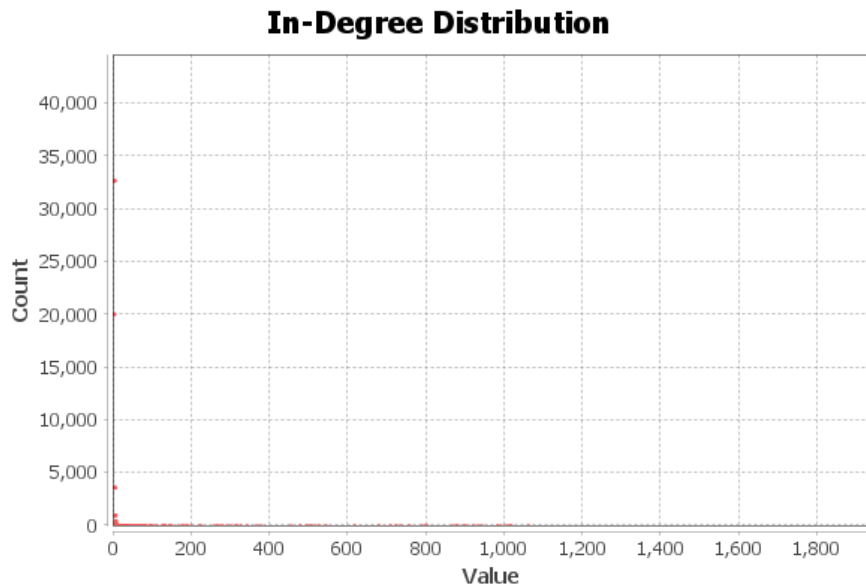


Figure 7: In-degree Distribution

The top 3 users in terms of in-degree are:

Label	Description	Weighted In-Degree	Weighted Out-Degree	Weighted Degree
stephenking.bsky.social	Writer	1059.0	2.0	1061.0
barnesandnoble.com	Book lover & whatnot BarnesandNoble.com	1016.0	54.0	1070.0
danielgreene.bsky.social	The Book Review Fantasy News Guy	1010.0	27.0	1037.0

Figure 8: Top Users in In-degree Distribution

These users are highly influential in the network and, while they are followed by lots of people, they do not follow many, which is why their out-degrees are miniscule compared to their in-degrees. The 1st one is @stephenking.bsky.social, a well-known horror books writer. The 2nd one is @barnesandnoble.com, a famous American bookseller company with the most retail outlets in the US. Lastly, the 3rd one is @danielgreene.bsky.social, a fantasy book writer who is well-known for his YouTube channel focused on book reviews and discussions.

Out-degree Distribution

Out-degree corresponds to the number of edges pointing outward from those vertices (Newman, 2018). Most nodes have a low out-degree, meaning they don't follow many other users. However, there could be a few nodes with high out-degree, which might be users who are very active in following, possibly bots or highly active users.

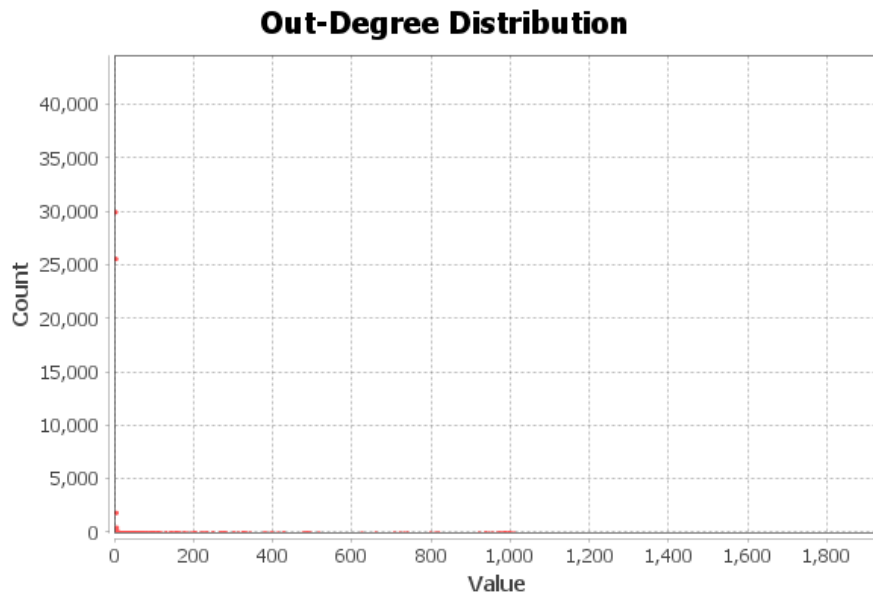


Figure 9: Out-degree Distribution

The top 3 users in terms of out-degree are:

Label	Description	Weighted In-Degree	Weighted Out-Degree	Weighted Degree
johnpearce650.bsky.social	Writer and environmentalist. Find me on Medium: http	929.0	1006.0	1935.0
emilycadwalader.bsky.social	MG & YA Fantasy Writer . she/her . same handle in var.	865.0	998.0	1863.0
robwinters.co.uk	Writer. My books robwinters.co.uk	499.0	996.0	1495.0

Figure 10: Top Users in Out-degree Distribution

All 3 of these users are, as can be seen from their descriptions, writers. Notably, the 1st user on this list, @johnpearce650.bsky.social, also ranks highest in overall degree. The 2nd user, @emilycadwalader.bsky.social, also has a relatively high degree and in-degree. The 3rd user, @robwinters.co.uk, however, is actively trying to climb in the network by following others, despite not having many followers in return, possibly indicating a strategic effort to increase visibility and engagement.

Centrality Measures

Betweenness Centrality

Betweenness centrality measures the extent to which a vertex lies on paths between other vertices (Newman, 2018). It identifies nodes that act as bridges or intermediaries in the network. In the network, most nodes have very low betweenness centrality, while a few nodes have much higher values. This is typical in social networks, where a small number of nodes (like key influencers or hubs) control the flow of information between different parts of the network. In fact, it turns out that almost all real-world networks have degree distributions with a tail of high-degree hubs like this, meaning it is right-skewed (Newman, 2018).

In this network, most nodes having low betweenness centrality suggests that most nodes are not critical for connecting different parts of the network. They are likely peripheral nodes that do not act as bridges between clusters or communities. In contrast, there are a few nodes with high betweenness centrality, which are crucial nodes that serve as intermediaries or bridges. They control the flow of information between different clusters in the network.

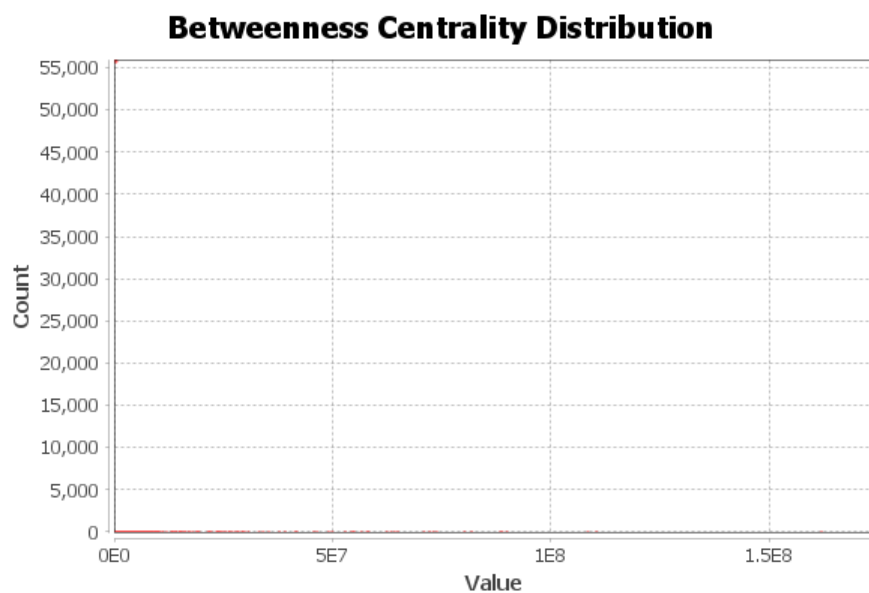


Figure 11: Betweenness Centrality Distribution

In order to better understand the aforementioned details, a visualization of the network can be made. The size of the nodes is determined by their betweenness centrality and the “ForceAtlas 2” layout is used.

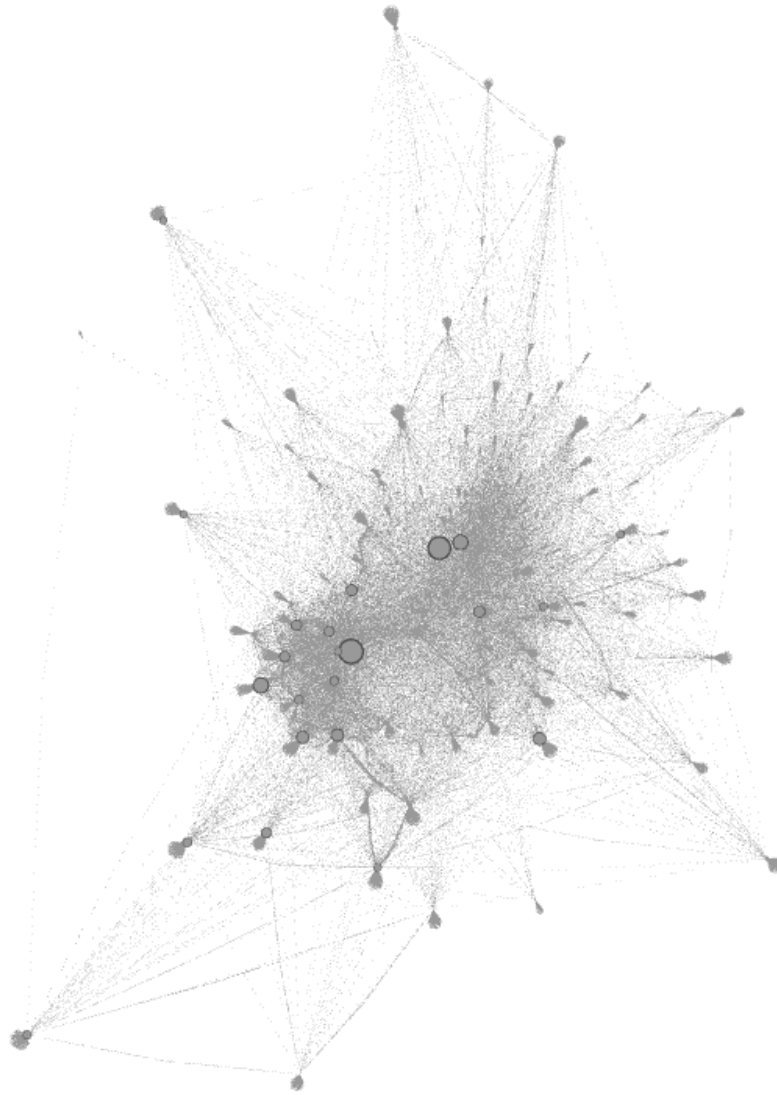


Figure 12: Betweenness Centrality Distribution in the Network

Additionally, partitioning the network using betweenness centrality and the “Fruchterman Reingold” layout, it is observed that approximately 95% of users have a betweenness centrality of 0, hence not being part of the influential users in the network.

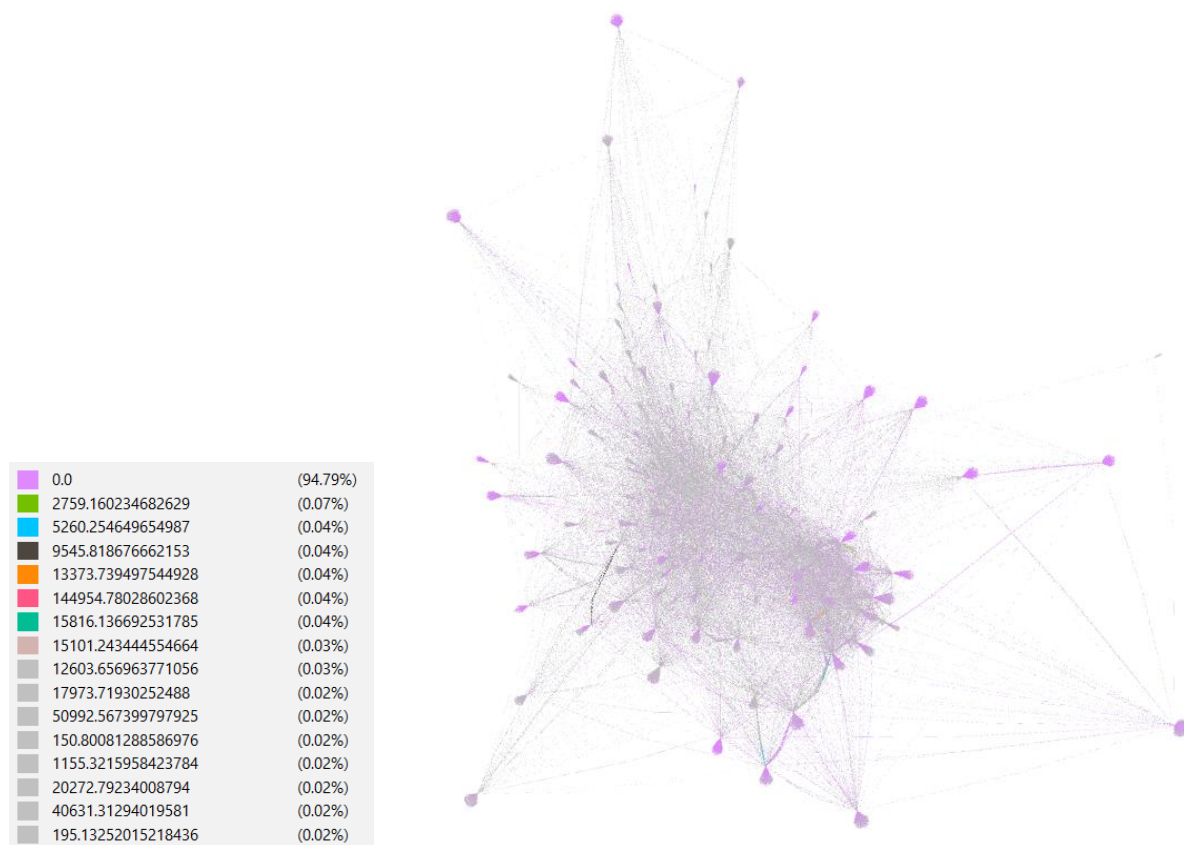


Figure 13: Partitioned Betweenness Centrality Distribution in the Network

The highest betweenness centrality in the network belongs to @christopherpaolini.bsky.social, which was expected given that the analysis is based on his network, while the 2nd highest to @rachelhartmanbooks.com, a YA author with an active account.

Closeness Centrality

Closeness centrality measures the mean distance from a vertex to other vertices (Newman, 2018). In the network, most nodes have low closeness centrality, indicating they are farther from other nodes, while a few have higher values, suggesting they are more central and can reach others more quickly. The nodes with low closeness centrality probably belong to the periphery or more isolated parts of the network, making them less central and slower in accessing or spreading information across the network, while the nodes with high closeness centrality are well-positioned to disseminate information efficiently.

The network seems to have a typical core-periphery structure, where a few central nodes (likely influential users or key figures such as @johnpearce650.bsky.social or @stephenking.bsky.social) play critical roles in connecting the network and spreading information, while many other nodes are more peripheral with limited influence. The high betweenness centrality of a few nodes implies that they are vital for maintaining

the network's cohesion. Their removal could fragment the network into disconnected components.

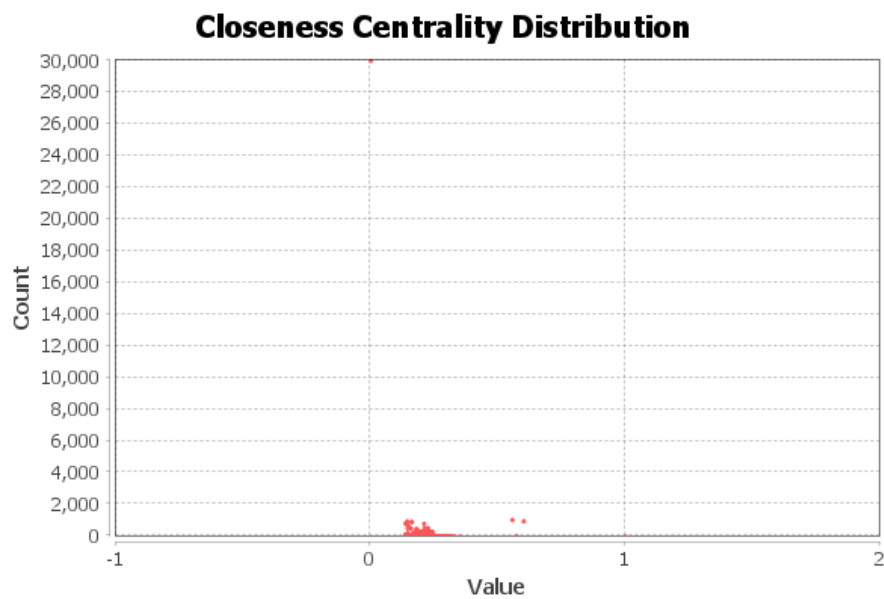


Figure 14: Closeness Centrality Distribution

To better understand the closeness centrality in the network, a visualization can be made to determine the size of the nodes by their closeness centrality. The layout used is "ForceAtlas 2".

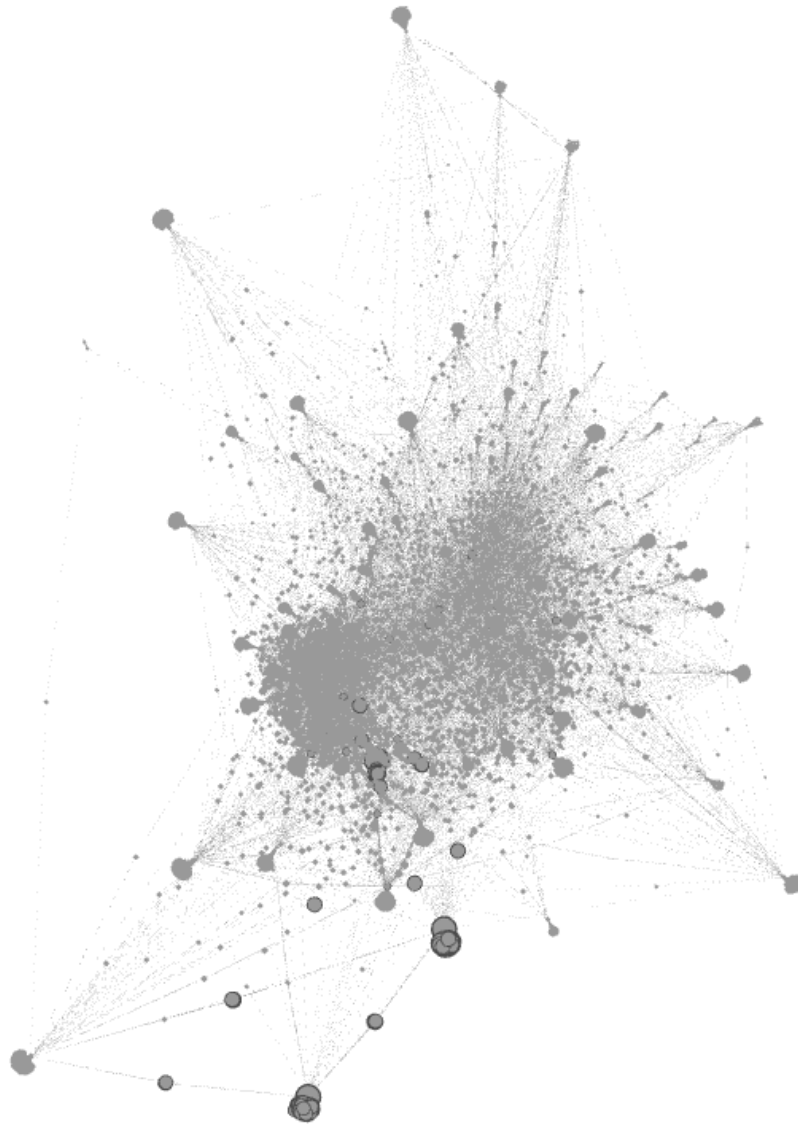


Figure 15: Closeness Centrality Distribution in the Network

Furthermore, by partitioning the network using closeness centrality and the “Fruchterman Reingold” layout, it is observed that approximately 51% of users have a closeness centrality of 0. This means that more than half of the users are far away from other users and are not easily accessible themselves. It is obvious that these nodes are located in the outer areas of the network, where groups have been created.

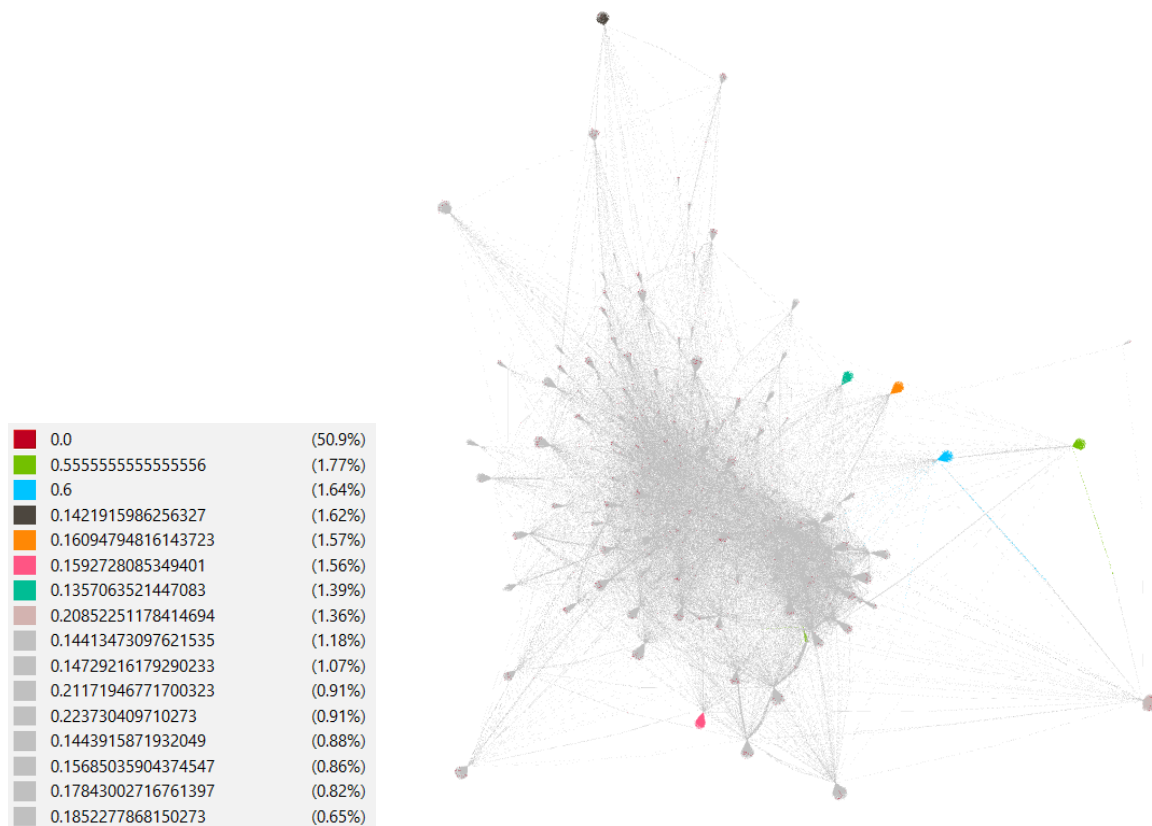


Figure 16: Partitioned Closeness Centrality Distribution in the Network

The highest closeness centrality in the network is equal to 1 and belongs to 3 users, @christopherpaolini.bsky.social, @filmandseries.bsky.social (an account devoted to Disney+) and @karenjbao.bsky.social (a writer with a relatively inactive account). All 3 of these users are positioned in the centre of the network and are highly connected to other users.

Eigenvector Centrality

Eigenvector centrality measures the influence that some nodes have in the network to other nodes. The eigenvector centrality of a node is defined to be proportional to the sum of the centralities of the node's neighbours, meaning that it takes into consideration the node's neighbourhood (Newman, 2018). In the network, the sum change is about 1,08 (more accurately 1,078211161341586). The few nodes with high eigenvector centrality are key figures or influencers who are not only well-connected themselves but also connected to other influential users. The distribution shows that a few nodes have significantly higher centrality, emphasizing the presence of central figures or hubs in the network.

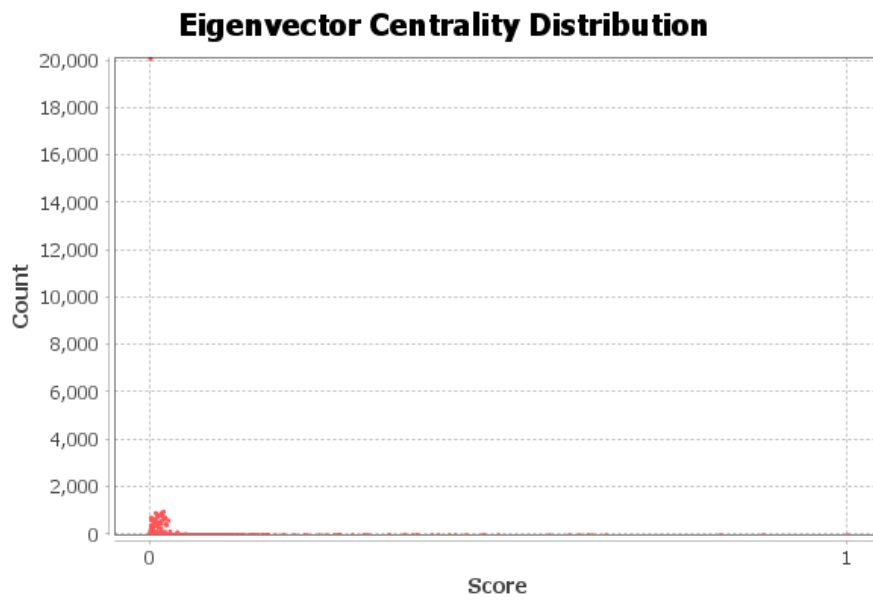


Figure 17: Eigenvector Centrality Distribution

To better understand the eigenvector centrality in the network, it can be visualized determining the size of the nodes by their eigenvector centrality and using the "ForceAtlas 2" layout.

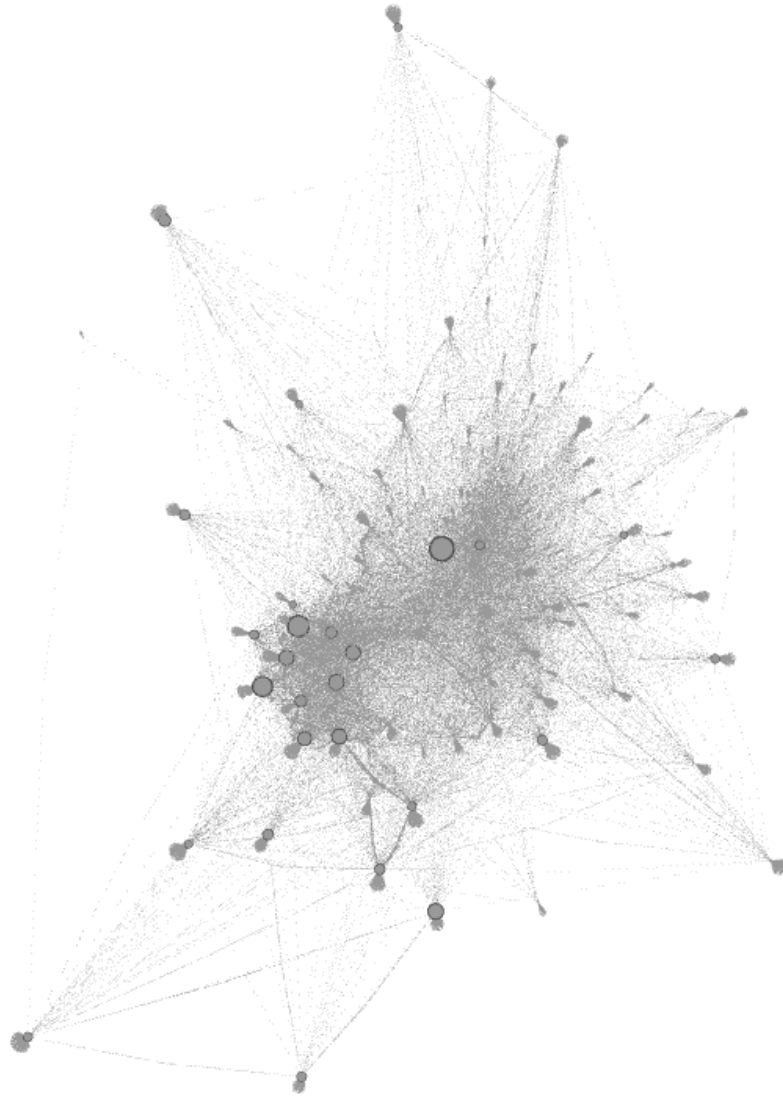


Figure 18: Eigenvector Centrality Distribution in the Network

In addition, by partitioning the network using eigenvector centrality and the “Fruchterman Reingold” layout, it is observed that approximately 34% of users have an eigenvector centrality of 0, while most of the other user’s centralities are also miniscule. Hence, a plethora of the users are not central themselves and are also not connected to other highly influential users.

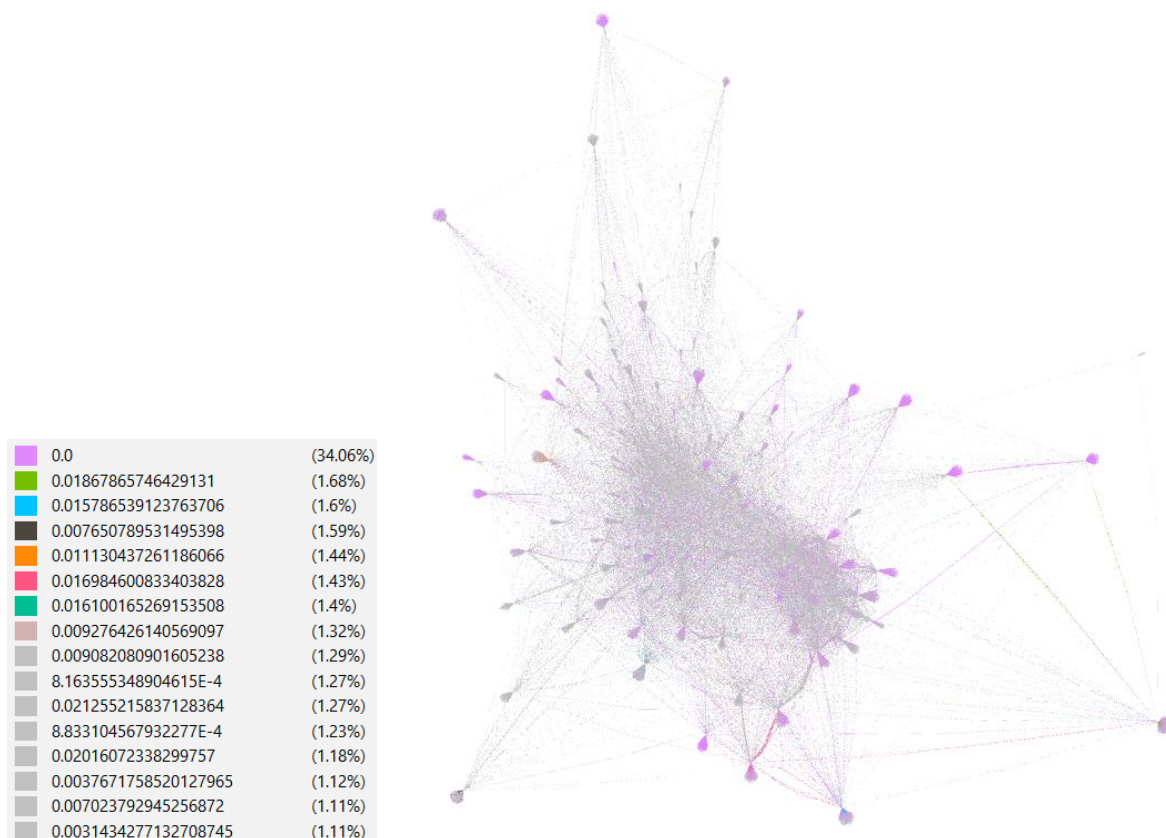


Figure 19: Partitioned Eigenvector Centrality Distribution in the Network

The highest eigenvector centrality in the network belongs to @christopherpaolini.bsky.social, as expected, while the 2nd highest to @lordtbrffa.bsky.social, the owner of FanFiAddict.com, a well-known book review blog.

Clustering Effects in the Network

The clustering coefficient captures the degree to which the neighbours of a given node link to each other (Barabási, 2016). In the network, the average clustering coefficient is 0,0009 and the value of clustering coefficient is approximately 0,0003 (specifically 3,193856682628393E-4). The very low clustering coefficient suggests that the network is sparsely connected. Specifically, it suggests that while there are some local clusters, most nodes are not part of tightly knit groups. Furthermore, the low value of clustering coefficient means that, on average, nodes have neighbours that are not well interconnected, reflecting a sparse and loosely connected structure for most of the network. In the network, this suggests that individual nodes typically do not form strong local clusters, and there are few groups where everyone knows everyone else.

Triadic closure is when an “open” triad of vertices (i.e., a triad in which one vertex is linked to the other two, but the third possible edge is absent) is “closed” by the addition of the last edge, forming a triangle (Newman, 2018). In the network, the number of triangles is 3.722 and the number of paths with length 2 is 34.960.867. The number of triangles suggests that there are some local clusters or tightly knit groups where users are mutually connected. However, given that this is a relatively small number of triangles compared to the size of the network, these tightly knit groups are not widespread. Additionally, the number of paths with length 2 is quite high, indicating that many nodes are indirectly connected through a shared connection, creating potential for triadic closure, though not all these paths lead to actual triangles, as seen in the relatively low number of triangles.

Partitioning the network by cluster using the layout “Force Atlas” and adjusting the nodes by size, it is visible that 97% of the users have a clustering coefficient equal to 0. Hence, most users are closely connected with their neighbours and have low chances of triadic closure, as discussed above. It must be noted that 0,34% of the users have a clustering coefficient equal to 1, meaning that a really small percentage of the networks is highly connected with the rest of the network.

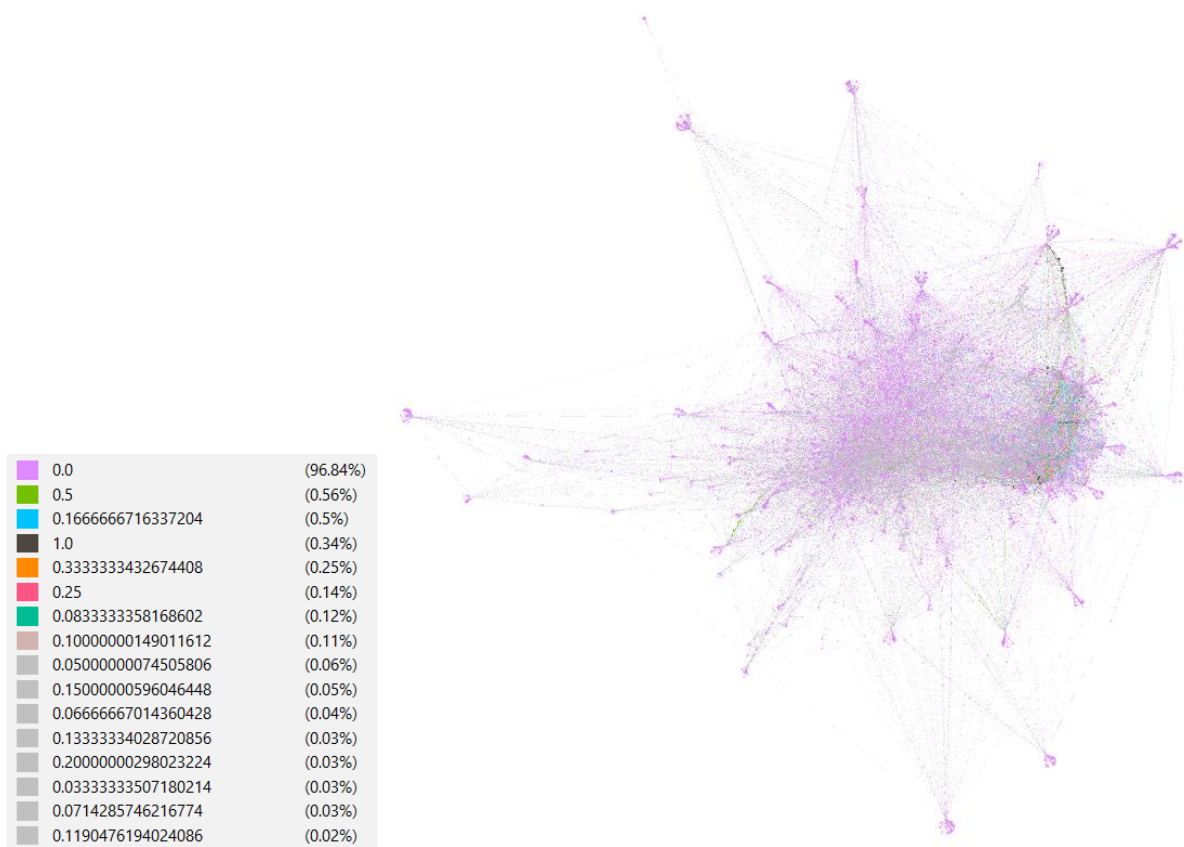


Figure 20: Partitioned Clustering Coefficient in the Network

Bridges and Local Bridges

Bridges provide nodes with access to parts of the network that are unreachable by other means (Easley & Kleinber, 2010). In the network, a few nodes have high bridging centrality and are key connectors between otherwise disconnected groups or clusters, while most have low bridging centrality. These play a crucial role in maintaining the network's cohesion, as expected based on previous findings. Nodes with high bridging centrality are likely users who connect different communities within the network. These bridging nodes help facilitate interaction between otherwise separate groups, promoting cross-community engagement.

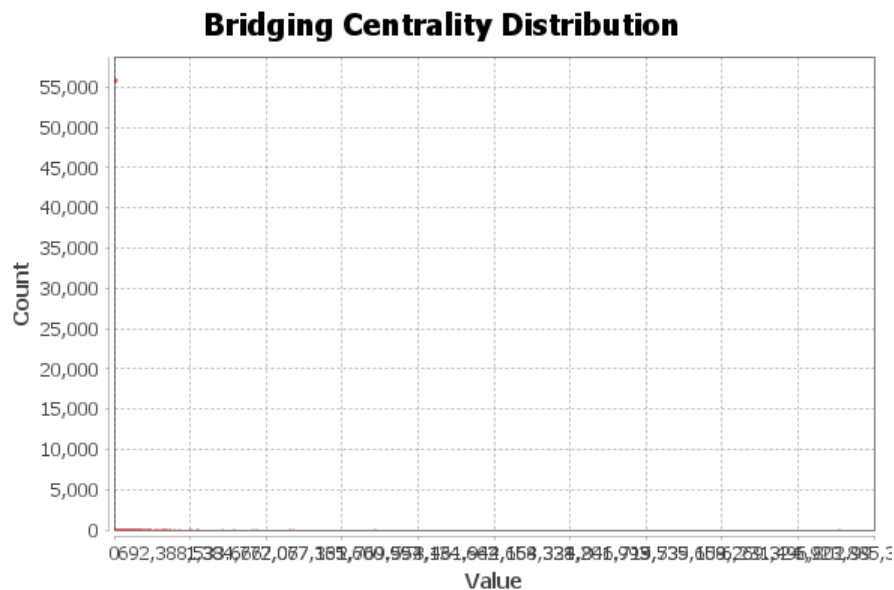


Figure 21: Bridging Centrality Distribution

To better understand the bridging centrality in the network, it can be visualized in a way that the size of the nodes is determined by their bridging centrality using the layout “ForceAtlas 2”.

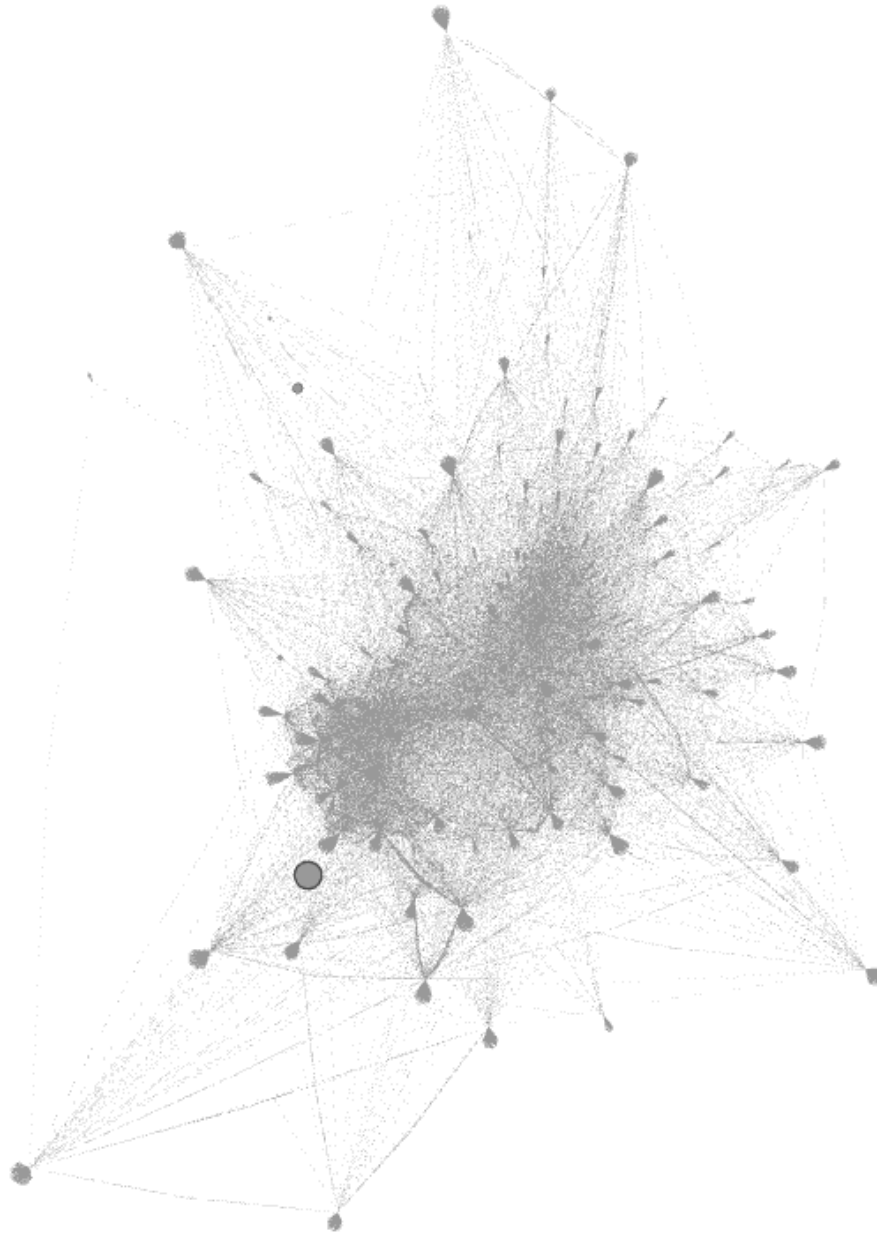


Figure 22: Bridging Centrality Distribution in the Network

In addition, by partitioning the network using bridging centrality and the “Fruchterman Reingold” layout, it is observed that approximately 95% of users have a bridging centrality equal to 0, while there are few people with extremely high centralities. This validates the fact that a few nodes facilitate interaction between otherwise unconnected groups, while most nodes are parts of isolated groups.

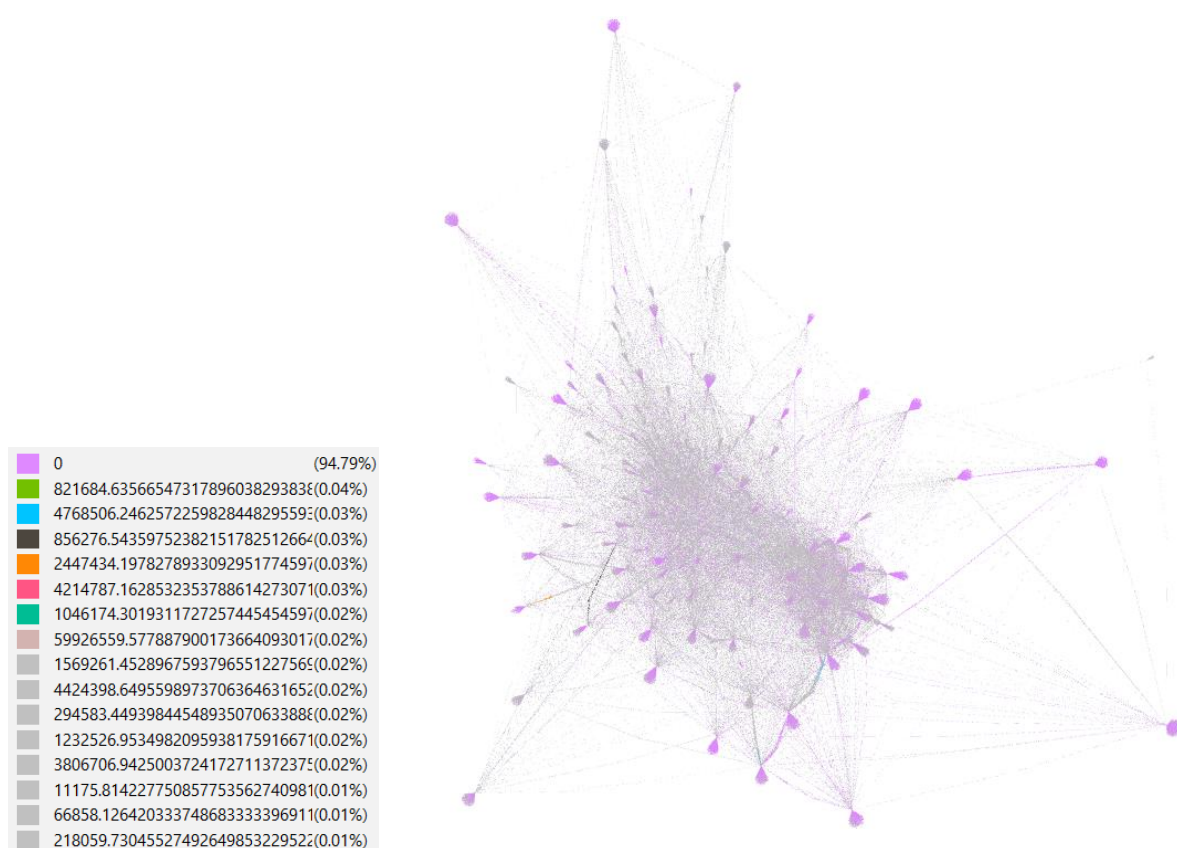


Figure 23: Partitioned Bridging Centrality Distribution in the Network

The highest bridging centrality in the network belongs to @jacqdoyle.bsky.social, a nonfiction and fiction writer and CNF editor, while the 2nd highest to @diceguy.bsky.social, who is the connection between @peaches.social and @moominmartin.bsky.social, 2 quite influential people in the network who are not in other ways connected.

Gender and Homophily

One of the most basic notions governing the structure of social networks is homophily, the principle that we tend to be similar to our friends (Easley & Kleinber, 2010). This means that similar nodes have higher probability to connect with each other than the nodes which are not similar (Newman, 2018).

To assess the homophily of the network, “Radial Axis” layout is used, nodes are grouped by modularity class and ranked by their degree.

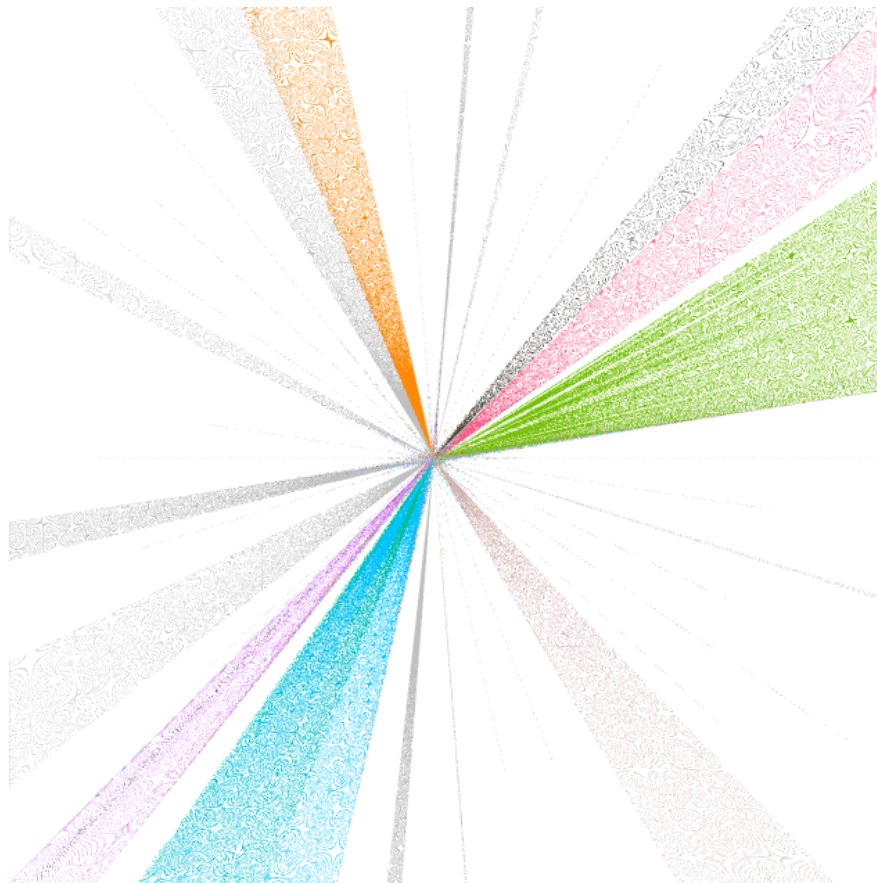


Figure 24: Homophily in the Network

The nodes that are closest to the centre are among the most important ones, while the ones radiating outward are connected to the respective central node. The distinct separation of colours indicates strong community structure within the network. This means there are clear boundaries between different communities, while the large wedges or groups of a single colour suggest they are tightly knit with many internal connections. Furthermore, the clear separation of communities indicates high homophily, where nodes with similar characteristics or interests are connected with each other.

Graph Density

The density of a graph is the fraction of the maximum possible number of edges and the actual number of edges (Newman, 2018). In the network, the density is extremely low, being almost 0. This means that the number of actual connections is very small compared to the total possible connections. This is typical for large social networks, where most nodes are connected to only a few others.

In the network, this low density suggests that it is sparse, with many potential connections not realized. Most users are not directly connected to most other users, and the network relies on a few key connections and nodes to maintain connectivity.

Community Structure

In the network, the modularity is 0,785. This is a relatively high value, indicating a better partition (Barabási, 2016). The network is well-structured into communities, with dense connections within these communities and sparser connections between them. The number of communities generated is 42. Hence, users tend to form distinct groups, which could be based on common interests, shared interactions, or following patterns.

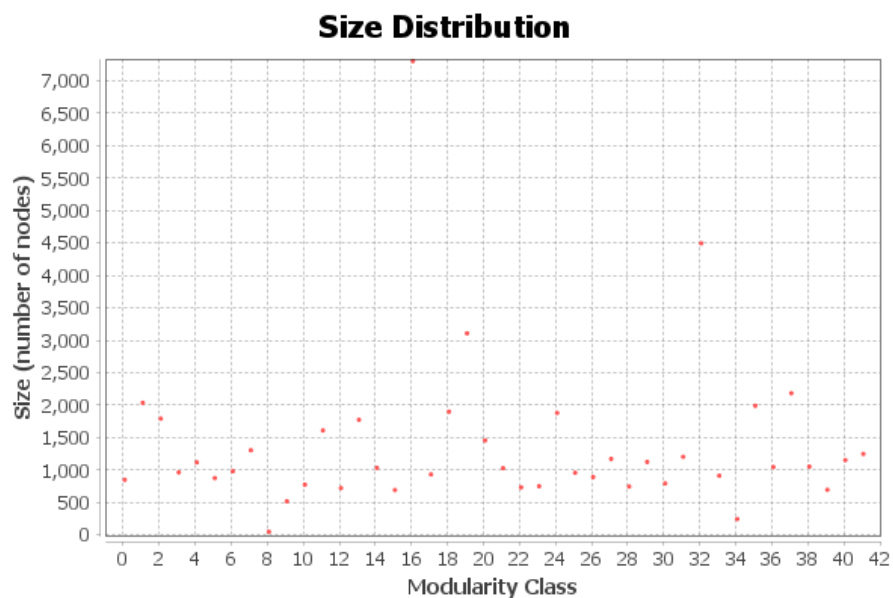


Figure 25: Community Size Distribution in the Network

These are the sizes, and the colours assigned to the top 12 communities, while the other 30 communities are coloured grey and have less than 2,5% of the nodes belonging to them. In this visualization the colour of the nodes depends on the community they belong to, and the used layout is “Fruchterman Reingold”.

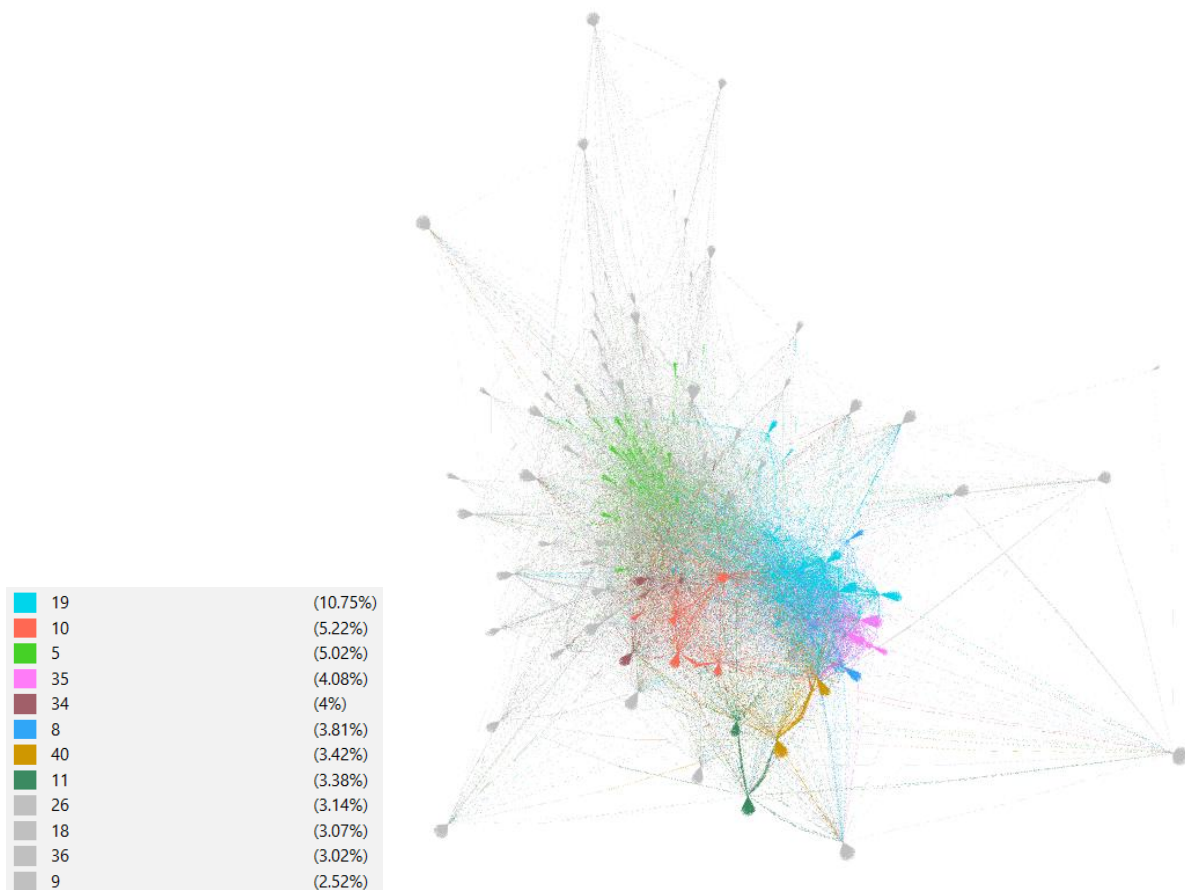


Figure 26: Partitioned Communities in the Network

The 1st community that is visible in the network consists of 6.346 nodes and 9.528 edges. This can be visualized by using the modularity filter and re-running the “Fruchterman Reingold” layout.

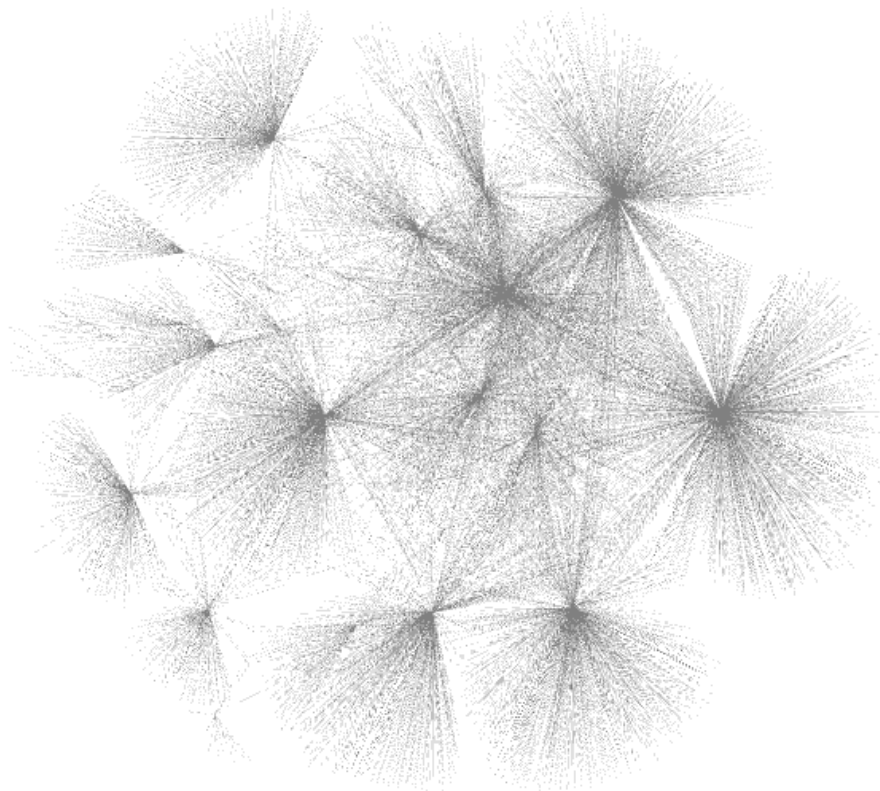


Figure 27: 1st Community in the Network

The account with the highest degree in this community is @lacrimamundi.bsky.social, who is the user with the 2nd highest weighted degree. The account with the highest betweenness centrality is @rachelhartmanbooks.com, who is a YA writer with an active account, as discussed. Lastly, the account with the highest PageRank and eigenvector centrality is @lordtbrffa.bsky.social, who surprisingly enough, connects with neither of the previous accounts, but, as mentioned, is the owner of FanFiAddict.com, a well-known book review blog.

The 2nd community that is visible in the network consists of 3.082 nodes and 3.818 edges. This can be visualized by using the modularity filter and re-running the “Fruchterman Reingold” layout.

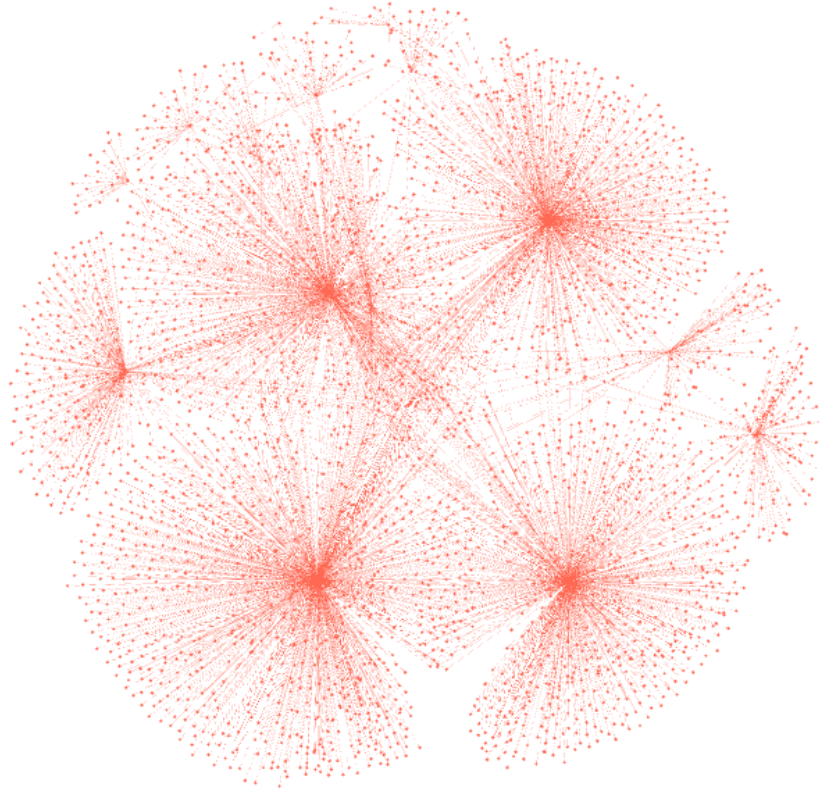


Figure 28: 2nd Community in the Network

The account with the highest degree in this community is @kmoore61.bsky.social, who has a multitude of followers, and the account with the highest closeness centrality and PageRank is @karenjbao.bsky.social, who, as mentioned, is a writer with a relatively inactive profile.

PageRank

Intuitively, we can think of PageRank as a kind of “fluid” that circulates through the network, passing from node to node across edges, and pooling at the nodes that are the most important (Easley & Kleinber, 2010). This centrality measure is commonly known as PageRank, which is the trade name given it by the Google web search corporation, which uses it as a central part of their web ranking technology (Newman, 2018). The Epsilon used for the analysis is 0,001 and the probability is 0,85. In the network, there are a few nodes with higher PageRank values, who have more significant or influential connections, not just in terms of quantity but also in the importance of the nodes they are connected to. They are likely key figures who are well-connected and whose connections are also to other important nodes.

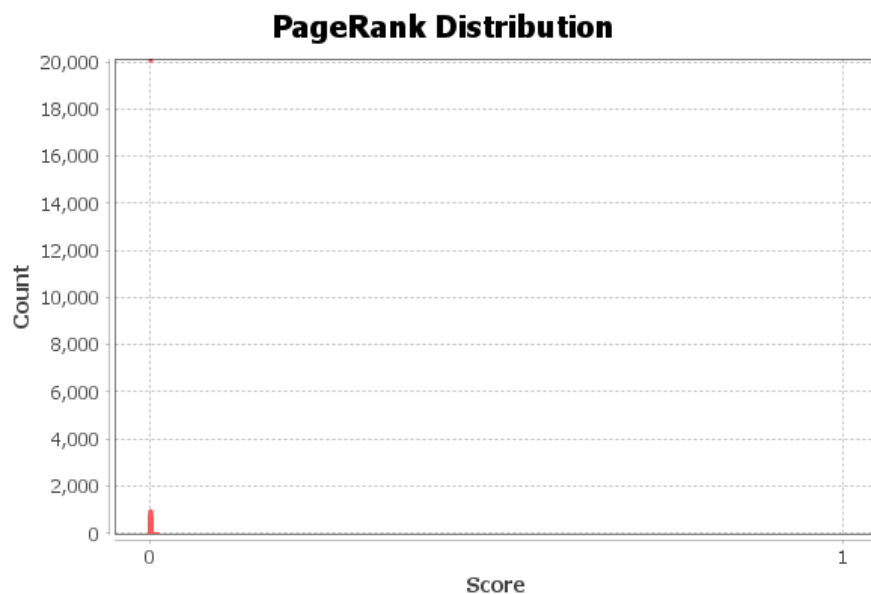


Figure 29: PageRank Distribution

To better understand the PageRank in the network, it is visualized partitioning the nodes based on their PageRank and using the layout “ForceAtlas 2”. It is visible that most users have a PageRank of approximately 0. These are users who do not have much influence in the network and do not contribute significantly to the overall structure or flow of information within it.

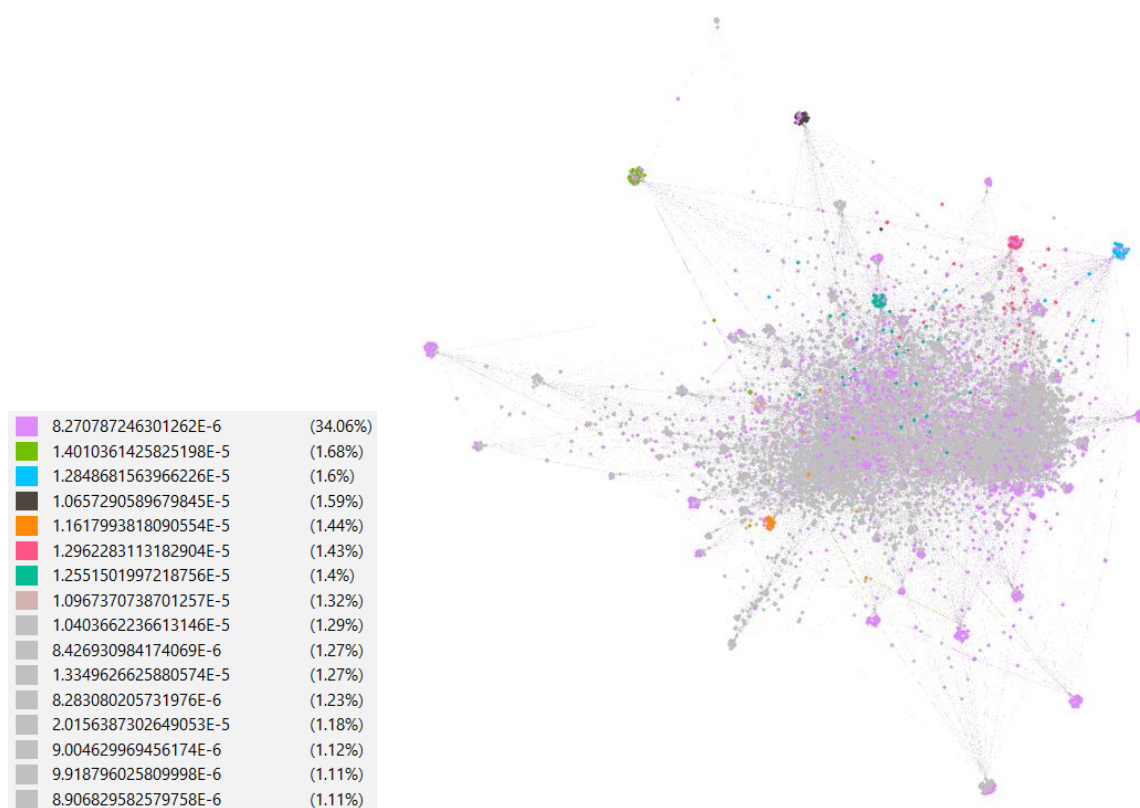


Figure 30: Partitioned PageRank Distribution in the Network

By filtering the data to include only users with a PageRank greater than 0,0001, 62 users were identified, representing the top 0,1% of the network. These users are highly influential within the network, warranting further analysis.

Among these accounts, @christopherpaolini.bsky.social stands out with the highest betweenness and eigenvector centrality, indicating a pivotal role in connecting various parts of the network and significant influence, @disneyplushelp.bsky.social has the highest bridging centrality, highlighting the importance of a movie streaming platform as a connector between different communities or clusters and @stephenking.bsky.social exhibits the highest closeness centrality, which is explained by the high in-degree of his account.

Interestingly, the highest PageRank in the network belongs to @peaches.social, a highly active account with 4.5K posts and numerous connections. Despite not being part of the writer community, @peaches.social plays a crucial role in the network's structure due to its activity and extensive connections.

Conclusion

The analysis of Christopher Paolini's Bluesky network has provided valuable insights into the structural and relational dynamics of his online community.

Key findings include the identification of central influencers such as @christopherpaolini.bsky.social, @rachelhartmanbooks.com, and @lordtbrffa.bsky.social, who hold significant influence within the network, as evidenced by their high betweenness and eigenvector centrality. These nodes play a pivotal role in maintaining the network's cohesion and facilitating information flow between different parts of the network.

The community structure of the network is well-defined, with a modularity of 0.785 indicating the presence of 42 distinct communities. This suggests that users within the network tend to form tight-knit groups based on common interests or interactions. However, the network's overall connectivity is sparse, with a low density and clustering coefficient. This indicates that while there are some clusters, most of the network is loosely connected.

Additionally, nodes with high bridging centrality, such as @jacqdoyle.bsky.social, are crucial in linking otherwise separate groups, enhancing the network's overall cohesion. These bridges facilitate interactions between different communities, promoting cross-community engagement and maintaining the network's integrity.

Lastly, an interesting insight is that an account can be a crucial part of the network without being particularly active or a member of a specific community. For example, @karenjbao.bsky.social, despite having a relatively inactive profile, holds a significant position within the network due to her high closeness centrality. Similarly, @peaches.social and @diceguy.bsky.social are not central figures in any specific community but serve as important connectors or influencers across different parts of the network. These findings emphasize that influence in a network is not solely determined by activity level or community membership but can also stem from strategic positions that enable effective communication and connectivity across the network.

This analysis provides a foundational understanding of how Paolini's digital social sphere is organized, and the role different nodes play in information dissemination and community formation.

However, the study is limited by the inherent constraints of data collection from a dynamic social platform like Bluesky. Future research could expand on this by incorporating longitudinal data to observe network evolution over time or by integrating content analysis to understand the nature of interactions. These steps could provide deeper insights into the evolving dynamics of Paolini's network and its influence within the broader literary community. This study serves as a step towards understanding the complex interplay of social media, influence, and community within the context of an author's digital presence.

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